

EX:- "I am going to" the model might predict words like "school", "market" or "work" based on previous usage frequency.

⇒ Various Grammar Based model:-

* Grammar-Based models use rules to generate or understand sentences in language.

These rules help computers figure out if a sentence is grammatically correct or how words relate to each other.

* Various computational grammars have been proposed and studied:-

- 1) Generative Grammars.
- 2) Hierarchical Grammar
- 3) Government and binding
- 4) Lexical Functional Grammar
- 5) Paninian Grammar - Based model.

① Generative Grammar

* It is a theory of grammar that holds that:

human language is shaped by a set of basic principles that are part of the human brain.

* Language is a relation between the sound and its meaning.

EX:- "The cat chased the mouse"

Rules define that the subject ("The cat") come before the verb ("chased").

(2) Hierarchical Grammar:-

* Chomsky categorized grammars into a hierarchical structure based on complexity. Hence,

→ type 0 (unrestricted grammar): Most powerful and complex. There is no restrictions and no rule.

→ type 1 (Context-sensitive grammar): Rules depending on the surrounding context.

→ type 2 (Context free grammar): Most commonly used in NLP, rules apply irrespective of context.

→ type 3 (Regular Grammar): Simplest grammar used in search engines and pattern recognition.

(3) Government and binding:-

* The theory explains how different parts of a sentence are related at various levels. It focuses on the structure and movement of words in sentence.

This theory assume 2 levels of existence. Sentences.

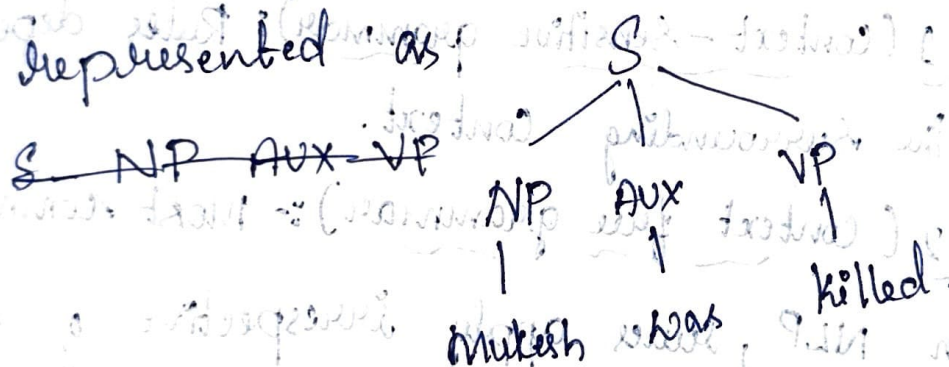
a) D-level (Deep Structure): Basic sentence meaning

b) S-level (Surface Structure): How the sentence is presented
and also identified two more levels of representation
called phonetic $\rightarrow$ Sound (How we spell) and

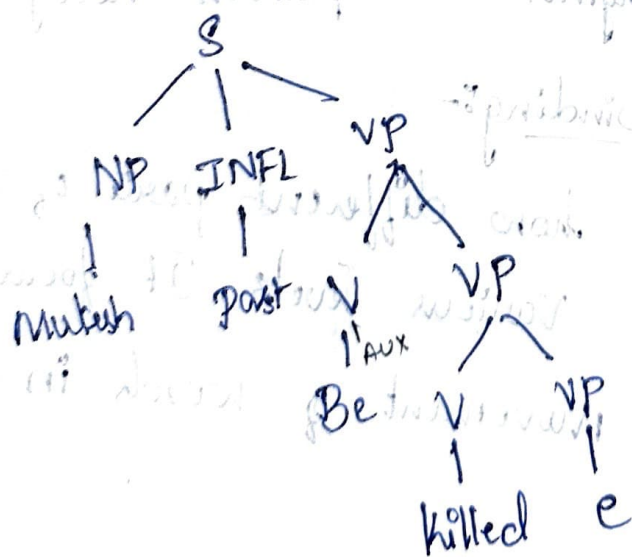
Logical Form (LF): meaning derived from sentence structure

Example: "Mukesh was killed"

(i) In ~~the~~ transformational grammar, this can be represented as:



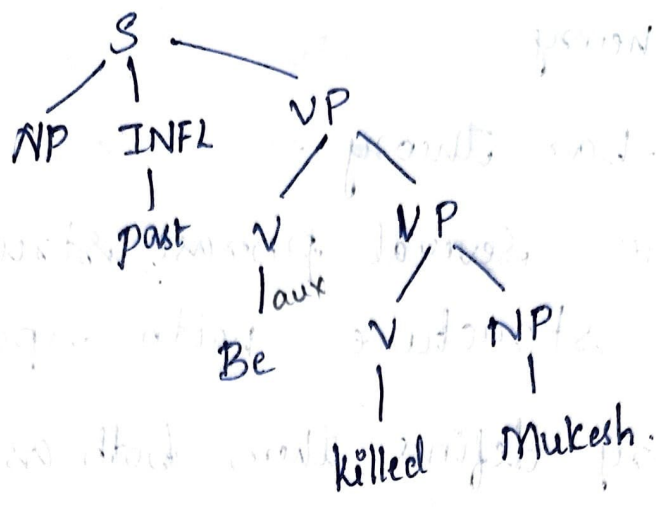
(ii) In GB, the S-level and D-level are as follows



(e) killed mukesh
(e) past kill mukesh

INFL: Inflection
NP: Noun Phrase
VP: verb phrase
e: empty

D-level

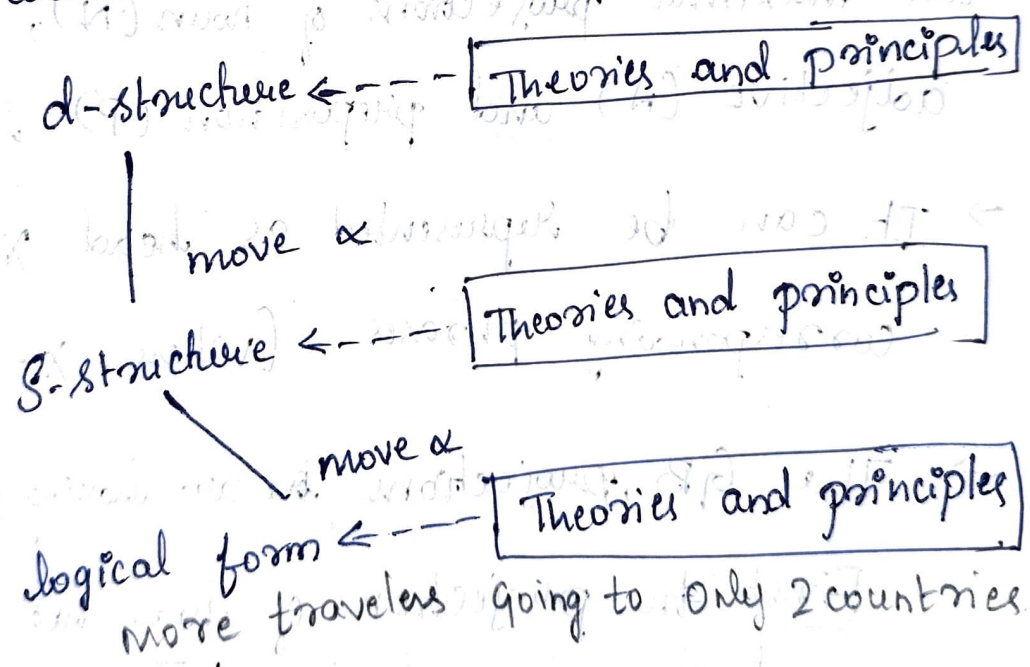


Components of GB :-

1) Move α (Move Anything Anywhere)

Ex:- "Most travelers visit two countries"

"Two countries are visited by most travelers"



2) Projection Principle:

Sentence structures must follow rules defined by the word from lexicon.

Ex:- Verbs need specific objects.
 open the door

3) X-bar Theory

→ It's X-bar theory.

It defines several phrase structures and the sentence structure with separate sets of rules,

→ $\bar{X}$ theory defines them both as maximal projections of some head.

→ Then the entities defined become language independent.

→ Thus, noun phrase (NP), verb phrase (VP), adjective phrase (AP), and prepositional phrase (PP) are maximal projections of noun (N), verb (V), adjective (A) and preposition (P) respectively.

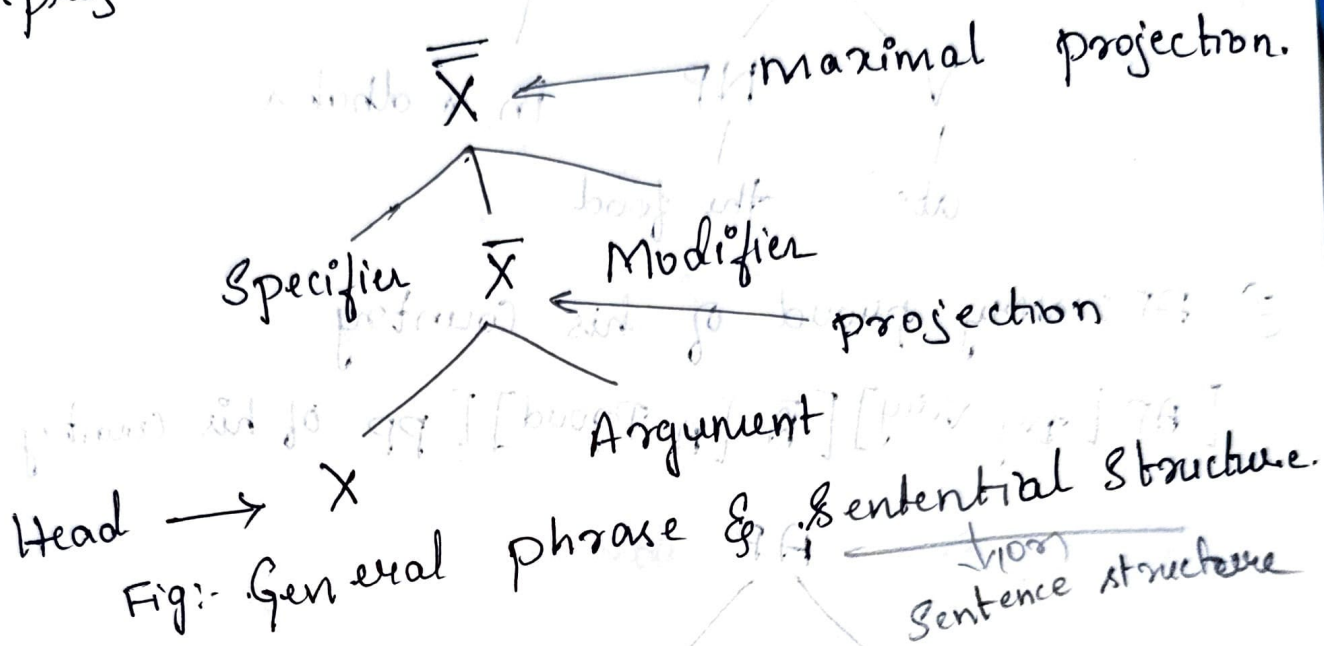
→ It can be represented as head X of their corresponding phrases (where, $X = \{N, V, A, P\}$)

→ The GB projections are having 2 levels:

First the projection of head at semi-phrase level, denoted by $\bar{X}$.

The second maximal projection, at the phrase level, denoted by $\bar{\bar{X}}$.

→ For sentences, the first level projection is denoted as S , and the second level is maximal projection is denoted by S' .



Let's consider,

① NP: the food in a dhaba

$[NP \text{ the } [N \text{ food}] PP [in a dhaba]]]$

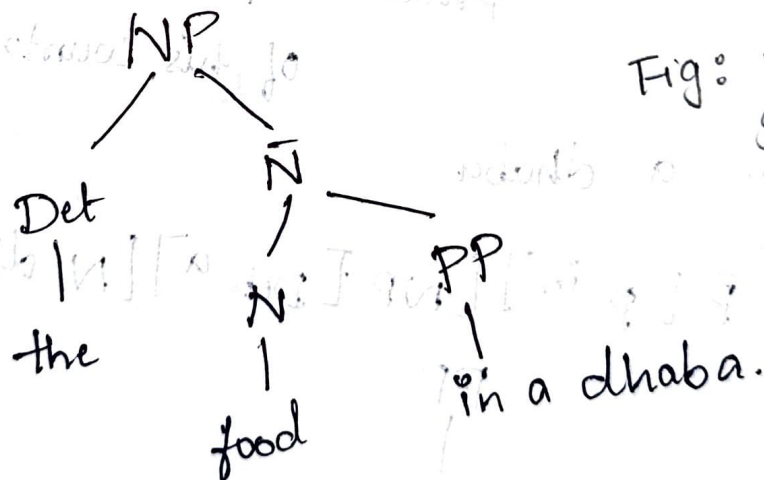
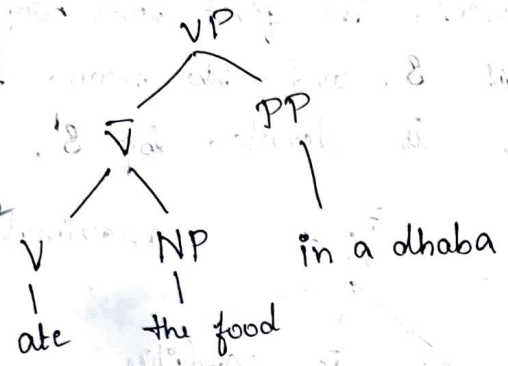


Fig: NP Structure

② VP: ate the food in a dhaba.

$[VP [V \text{ ate}] [NP \text{ the food}] [PP \text{ in a dhaba}]]]$

Fig:- VP Structure



③ AP: Very proud of his country

[AP [Deg very] [Ā [A Proud] [PP of his country]]]

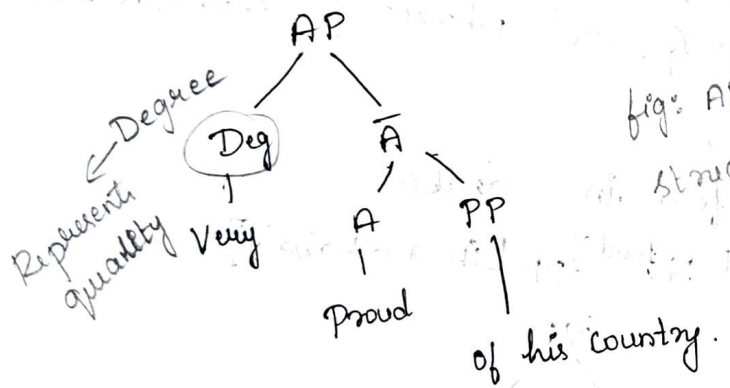


fig: AP Structure

④ PP: in a dhaba

[PP [P̄ [P in] [NP [Det a] [N dhaba]]]]

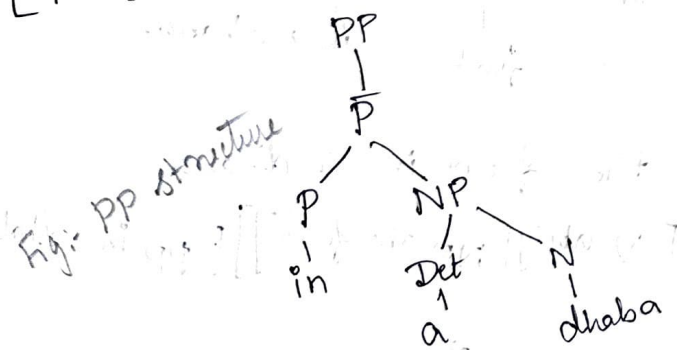
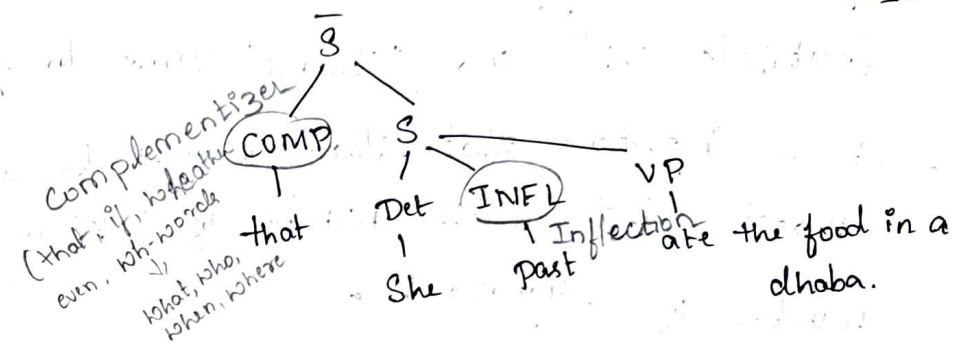


Fig:- PP structure

⑤ S: that she ate the food in a dhaba.
[S [comp that] [S [Det she] [INFL Past] [VP ate the food in a dhaba]]]



4) Theta (θ) theory:-

Assigns roles to words.

Ex:- 'Mukesh ate an apple'

(Mukesh = Agent, Apple = Theme).

Theta-role & Theta criterion.

Ex:- The verb 'eat' can take arguments with

θ-roles, 'Agent, Theme'.

Agent is a special type of role which can be assigned by a head to outside arguments (external arguments), whereas the other role are assigned within its domain (internal arguments).

Hence, "Mukesh ate food" the verb 'eat' assigns the "Agent" role to 'Mukesh'

(outside VP) and "Theme" role to 'food'

- It is important that there should be a match between the number of roles and number of arguments.

5) Binding theory

Defines noun-pronoun relationships.

EX: a) "Mukesh knows himself". (correct) vs

b) "Mukesh believes that Amrita knows him" (pronoun)

c) "Mukesh knows him". (Ambiguous)

believes that Amrita knows Nupur

Binding theory can be given as:-

a word or phrase that refers back to an earlier word or phrase

a) An anaphor (+a) is bound in its governing category.

EX:- My cousin said (she) was coming

b) A pronominal (+p) is free in its governing category.

relating to pronouns or like a pronoun EX:- "My Book"

c) An R-expression (-a, -p) is free. "This dog"

Referring expression EX:- The cat sat on the mat

"The cat" Refers to a specific feline

"The mat" Refers to a specific piece of floor covering

⑤ Empty Categories:-

Invisible placeholders created when words move.

EX:- "What did Mukesh eat?" ("What" leaves a trace)

⑥ Sub categorization

Filters to control which words can follow a specific word.

EX:-

The verb "eat" can sub-categorize for NP,

whereas the verb "Sleep" cannot.

Hence "eat food" is well formed, but

"Sleep the bed" is not.

⑦ C-command:-

It defines the scope of a word within a sentence.

⑧ Government:-

A word controls another within specific boundaries

α governs Biff:

α C-commands B

α can be head (Noun, Verb, preposition, adjective, inflection) etc

⑨ movement : Move α described as "move anything anywhere"; through it provides restrictions for valid movements.

Another movement wh-movement, a wh-phrase is moved as follows.

What did Mukesh eat?

[Mukesh INFL eat what]:

⑩ Bounding and Control Theory

① Bounding Theory

* Explains : long-distance movement of elements in a sentence

* In English, when a complement clause (a clause that completes the meaning of a sentence) undergoes movement, it follows specific constraints

* The movement is bounded if NP (noun phrase) and S_i function as bounding nodes

* The rule: move α (any syntactic movement) cannot cross more than one bounding node at a time.

⑥ Control Theory

- * Deals with how subjects are interpreted when they are not explicitly mentioned.
- * It involves syntax, semantics, and pragmatics to determine meaning.
- * The empty category PRO behaves differently based on context.
- * It acts like an anaphor (referring to another element in the sentence) when it is the subject of a complement clause with verbs like decide and try.
- * It behaves like a pronoun with other verbs.

⑦ Lexical functional Grammar (LFG) Model

- * Lexical functional Grammar is a formal system for Grammatical Representation.
- * It was developed by Kaplan and Bresnan in 1982.
- * Unlike GB, LFG represents sentences at two syntactic levels: constituent structure (C-structure) and functional structure (f-structure).

* The LFG is a formalism that is both computationally and linguistically motivated and provides precise algorithms for linguistic issues it can handle.

* Lexical functional is composed of two terms: the 'functional' part is derived from

'grammatical functions', such as subject and object, (or) roles played by various arguments in a sentence.

* Constituent Structure (C-structure)

* Represents the hierarchical arrangement of words in a sentence

* Defines how words and phrases combine.

* Uses Context-free grammar (CFG) rules.

$S \rightarrow NP VP$

$VP \rightarrow V \{NP\} \{NP\} PP^* \{S\}$

$PP \rightarrow P NP$

$NP \rightarrow Det N \{PP\}$

$S' \rightarrow Comp S$

$S \rightarrow$ Sentence

$S' \rightarrow$ clause.

$\Rightarrow$ Functional Structure (f-structure)

* Captures grammatical functions like subject, object, predicate, and adjuncts.

* Represents abstract relationships and attributes such as tense, case, and agreement.

Ex:- "She saw stars in the sky"

C-structure analysis

CFG Rule.

$S \rightarrow NP VP$

$NP \rightarrow$ Pronoun

$VP \rightarrow V NP PP$

$PP \rightarrow P NP$

F-structure Analysis

Attribute	Value
Subject	She
Predicate	Saw
Object	Stars
Oblique	in the sky

Rule-1: $S \rightarrow NP VP$
 $\uparrow$ Subj = 1 $\uparrow$ A = 1

T Subj = 1

Subject is in position

(the NP)

Tree transformation/labeling implying the sentence

a whole corresponds to transformation

Rule-1.

Lexical entries of various words in the sentence.
 g: She saw stars

Lexical entries → "She", "saw", "stars".
Explanation:-

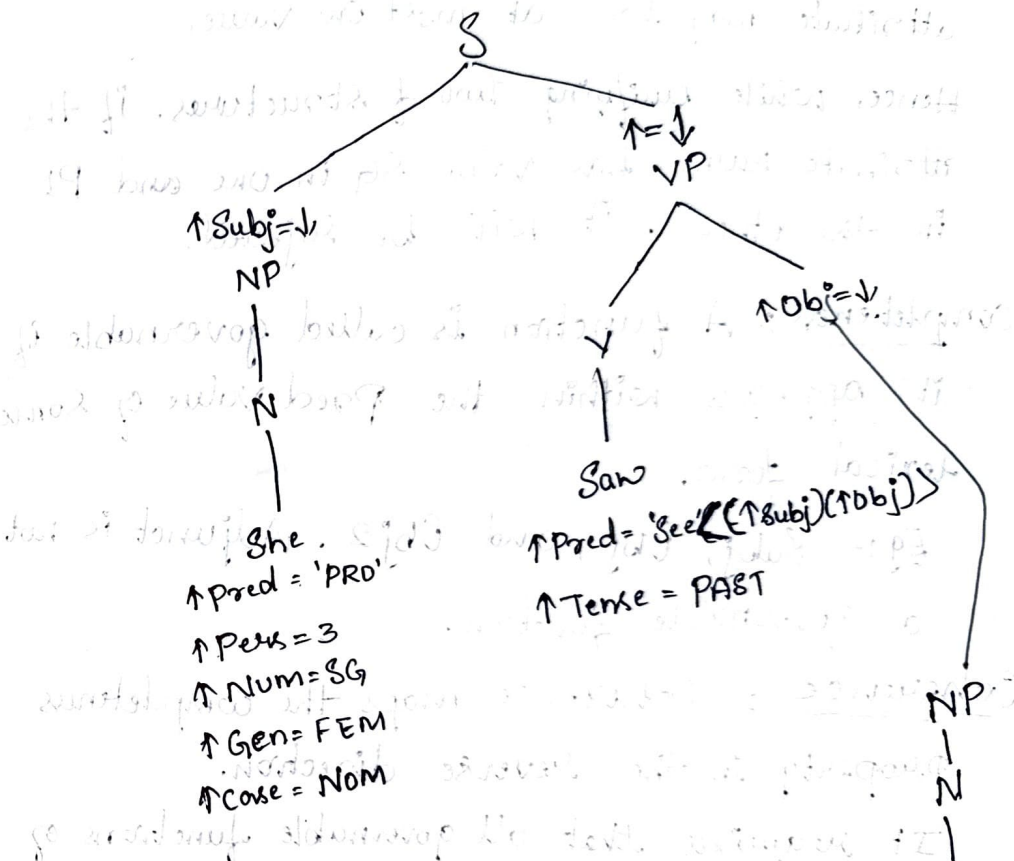
① She (N) ($\uparrow$ pred) = 'PRO' → It is a pronoun
 ($\uparrow$ pers) = 3 → It is a third person
 ($\uparrow$ Num) = SG → Singular
 ($\uparrow$ Gen) = FEM → Feminine or Female
 ($\uparrow$ case) = NOM → Nominative case.

② Saw (V) $\uparrow$ pred = 'See <($\uparrow$ subj)($\uparrow$ obj)>'
 This is the verb 'see' in the past tense.

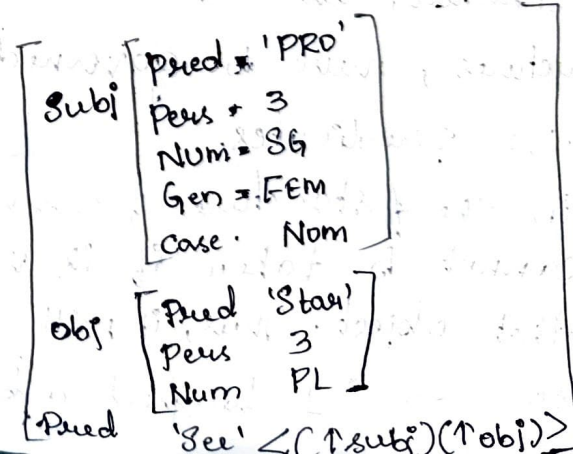
Its predicate argument structure
 ($\uparrow$ pred) = 'See <($\uparrow$ subj)($\uparrow$ obj)>'
 indicates that it takes a subject and an object.

③ Stars (N): $\uparrow$ pred = 'Star' → pred (Noun)
 $\uparrow$ pers = 3 → 3rd person
 $\uparrow$ Num = PL → plural.

C-Structure of Sentence



f-Structure



$\uparrow$ pred = 'Star'
 $\uparrow$ pers = 3
 $\uparrow$ Num = PL

Consistency :- In a given f-structure, a particular attribute may have at most one value.

Hence, while unifying two f-structures, if the attribute Num. has value SG in one and PL in the other, it will be rejected.

Completeness :- A function is called governable if it appears within the Pred. value of some lexical form.

Eg:- Subj, Obj, and Obj2. Adjunct is not a governable function.

Coherence :- Coherence maps the completeness property in the reverse direction.

It requires that all governable functions of an f-structure, and all its subsidiary f-structures, must be governed by their respective predicates.

Hence, in the f-structure of a sentence, an object cannot be taken if its verb does not allow that object. Thus, it will reject the sentence → "I laughed a book".

Consistency :-

* Attributes must and should have a single value

Ex:- Correct "He eats an apple".

Incorrect "He eat an apple" (Subject-verb agreement violation)

Completeness :-

⇒ A sentence must contain all required elements defined by its predicate.

Ex:- Correct: "He saw a movie".

Incorrect: "He saw" (missing object)

⇒ Coherence :-

Functions must align with the grammar role defined by the predicate.

Ex:- Correct: "I ate dinner"

Incorrect: "I laughed a book"

(Laugh doesn't take an object)

→

⇒ Lexical Rules in LFG :- ⇒ Active and passive constructions :-

* Active sentence example:

"Tara ate the food"

Predicate: eat

Structure: Subject = Tara,
object = the food

Representation = $\uparrow \text{pred} = \text{'eat'} \langle (\uparrow \text{subj}) (\uparrow \text{obj}) \rangle$

* Passive sentence example:

"The food was eaten by Tara"

Predicate: eat

Structure: Subject = food, Oblique Agent = Tara

Representation = $\uparrow \text{pred} = \text{'eat'} \langle (\uparrow \text{oblg}) (\uparrow \text{subj}) \rangle$

⇒ Paninian Grammar - Based Model :-

- * It was written by Panini in 500BC in Sanskrit
- * The framework can be used for Indian languages and possibly some Asian language.
- * Unlike English (SVO), most Indian and Asian languages follow SOV order.

Features of Indian Languages

Indian languages emphasize oral communication for knowledge propagation.

The primary goal is to communicate ideas effectively.

Ex: ① "माँ बच्चे को खाना देती है।

Mother child to food give (S)
Mother gives food to the child.

② उसने खाना खाया

He (Subj) food ate
He ate food

⇒ Karaka Theory

- * Karaka means case relations in Paninian Grammar
- * Different Karaka relations:
 - Karaka (Subject) - The doer of the action
 - Karma (Object) - The entity affected by the action
 - Karana (Instrument) - The means by which an action is performed.
 - Sampradana (Beneficiary) - The recipient of the action.
 - Apradan (Separation) - The source or point of departure.
 - Adhikaran (Locus) - The location where the action happens.

Ex:- माँ बच्ची को आँगन में हाथ से रोटी
खिलाती है ॥

Mother child to courtyard-in hand-by
bread feed (s)

Mother feeds bread to the child by hand in the courtyard.

⇒ Layered Representation in PG:

It includes multiple levels:

- * Semantic level: Meaning and Context
- * Karaka level: Role of participants in an activity
- * Vibhakti level: Based on inflections or grammatical relations
- * Surface level: Basic sentence structure.

⇒ Issues in Paninian Grammar

* Computational Issues in PG

→ Computational Implementation of PG

• Complexity in parsing and analyzing inflectional variations.

• Need for efficient algorithms to process language structures.

→ Adaptation of PG to Indian and similar languages:

- Variability in grammar and regional language differences.
- Need for extensive linguistic resources and rule-based frameworks.

* Paninian Grammar provides a robust framework for Indian and Asian languages.

* Karaka theory effectively represents sentence roles.

⇒ Statistical Language Model.

* The model assigns probabilities to sequences of words.

* These models are crucial for applications like speech recognition, machine translation, and text prediction.

* One of the most widely used and effective approaches in the N-gram model, which

Simplifies the task of computing sentence probabilities by breaking down the problem into smaller, manageable parts.

(a) n-gram model

* It estimates the probability of a sequence of words by considering the occurrence of N words together.

* It decomposes the probability of a sentence using conditional probability.

Formula,

$$P(S) = P(w_1, w_2, w_3, \dots, w_n)$$

$$= P(w_1) \cdot P(w_2 | w_1) \cdot P(w_3 | w_1, w_2) \dots$$

$$= \prod_{i=1}^n P(w_i | h_i)$$

* It calculates the probability of each word based on the previous ones

So, in order to calculate sentence probability we need to calculate the probability of a word.

$$P(w_i | h_i) \approx P(w_i | w_{i-n+1} \dots w_{i-1})$$

w_i = No. of words
 h_i = History of words

Ex:- Consider 2 sentences

"There was heavy rain" $\rightarrow$ It occurs many times
"There was heavy flood" $\rightarrow$ It doesn't occur many times.

Different types of n-grams

* Unigram: Considers, each word independently ignoring previous words.

$$P(w_i)$$

Ex:- The probability of the word "Rain" occurring with out any context.

* Bigram: Considers only the previous word.

$$P(w_i | w_{i-1})$$

Ex:- The probability of the word "rain", given the previous word "heavy".

* Trigram: Considers the 2 previous words.

$$P(w_i | w_{i-2}, w_{i-1})$$

* n-gram: Considers $N-1$ previous words.

Ex:- To find the probability of "There was heavy rain"

$$\begin{aligned} * P(S) &= P('There', 'was', 'heavy', 'rain') = P \\ &= P('There') \cdot P('was' | 'There'). \end{aligned}$$

IMPORTANT NOTES

$P('heavy' / 'There\ was') \cdot P('rain' / 'There\ was\ heavy')$

N-gram Model Example

Because it's hard to calculate probabilities of large contexts using Markov assumption.

We often simply use Bi-gram model,

To calculate the probability of,
 $P^*(\text{'There was heavy rain'}) \sim P(\text{'There'}) \times P(\text{'was' | 'There'}) \times P(\text{'heavy' | 'was'}) \times P(\text{'rain' | 'heavy'})$.

⇒ Smoothing Techniques:-

- * This in an n-gram model are used to handle the issue of zero probability for unseen n-grams to adjust probability estimates.
- * If an unseen n-gram appears in new data, the model would assign it a probability of zero, which is problematic for many applications like speech recognition and machine translation.

① Add-One Smoothing

② Good-Turing Smoothing

→ Add-One Smoothing:-

This is a simple technique where 1 is added to the count of every N-gram before converting them into probabilities.

IMPORTANT NOTES

$$P(w_i | w_{i-1}) = \frac{C(w_{i-1} w_i) + 1}{C(w_{i-1}) + V}$$

Where :-

* $C(w_{i-1} w_i)$ = count of the bigram $w_{i-1} w_i$ in the training corpus.

* $C(w_{i-1})$ = count of the unigram w_{i-1}

* V = Vocabulary size (number of unique words)

* $+1$ = The Smoothing factor.

EX:-

I	love	dogs	}	love = 2
I	love	cats		dogs = 1
				Cats = 1

$$\frac{2+1}{1+1}$$

→ Good - Tuning Smoothing

It redistributes probability mass from frequent events to rare or unseen events.

This adjusted frequency f^* is calculated as,

$$f^* = (f + 1) \frac{n_f + 1}{n_f}$$

Where,

f = original frequency of the N-gram

n_f = number of N-grams that occur exactly f times

n_{f+1} = no. of N-grams that occur exactly $f+1$ times

EX:- If there are 2 bigrams that occurring 5 times
~~f~~ = (B+1) ($n_5=2$) and 3 bigrams
 occurs 6 times ($n_6=3$).

To compute f^* for n -gram occurring 5 times

$$f^* = (5+1) \times \frac{3}{2} = (6) \times \frac{3}{2} = \frac{18}{2} = 9$$

EX:- Training set:- 1) The Arabian knights
 2) These are the fairy tales of the east
 3) The stories of the Arabian knights are translated
 in many languages.

So:- The arabian knights are the ^{fairy tales} stories of the east.

$$P(\text{The} | \langle s \rangle) \times P(\text{Arabian} | \text{the}) \times P(\text{knights} | \text{Arabian}) \times P(\text{are} | \text{knights}) \\ \times P(\text{the} | \text{are}) \times P(\text{fairy} | \text{the}) \times P(\text{tales} | \text{fairy}) \times P(\text{of} | \text{tales}) \\ \times P(\text{the} | \text{of}) \times P(\text{east} | \text{the}).$$

$$= 0.67 \times 0.5 \times 1.0 \times 1.0 \times 0.5 \times 0.2 \times 1.0 \times 1.0 \times 1.0 \\ \times 0.2$$

$$= 0.0067$$